

Assignment 3

Digital Signal Analysis
Faculty Name: Dr. Chiranjeevi Yarra

Due date: 08:00am on 22nd April 2026

1. Solve all the question and submit a handwritten document.
2. Plagiarism will be penalised.
3. You can use any coding/programming language to complete ‘Coding’ questions.
4. Submit a pdf of the form <roll no>_dsa3.pdf
5. The submission should be a zip folder named with your rollnumber-dsa . Each coding question must be placed in a separate folder containing the corresponding code and results.

Part I

Theory Questions

Question 1

The following input-output pairs have been observed during the operation of a **time-invariant** system:

$$x_1(n) = \{ \underset{\uparrow}{1}, 0, 2 \} \xrightarrow{T} y_1(n) = \{ \underset{\uparrow}{0}, 1, 2 \}$$

$$x_2(n) = \{ 0, \underset{\uparrow}{0}, 3 \} \xrightarrow{T} y_2(n) = \{ \underset{\uparrow}{0}, 1, 0, 2 \}$$

$$x_3(n) = \{ 0, 0, 0, \underset{\uparrow}{1} \} \xrightarrow{T} y_3(n) = \{ 1, \underset{\uparrow}{2}, 1 \}$$

Can you draw any conclusions regarding the **linearity** of the system? What is the **impulse response** of the system?

Question 2

The following input–output pairs have been observed during the operation of a **linear** system:

$$x_1(n) = \{-1, \underset{\uparrow}{2}, 1\} \xrightarrow{T} y_1(n) = \{1, \underset{\uparrow}{2}, -1, 0, 1\}$$

$$x_2(n) = \{1, \underset{\uparrow}{-1}, -1\} \xrightarrow{T} y_2(n) = \{-1, \underset{\uparrow}{1}, 0, 2\}$$

$$x_3(n) = \{0, \underset{\uparrow}{1}, 1\} \xrightarrow{T} y_3(n) = \{\underset{\uparrow}{1}, 2, 1\}$$

Can you draw any conclusions about the **time invariance** of this system?

Question 3

Consider the following three operations.

- (a) Multiply the integer numbers: 131 and 122.
- (b) Compute the convolution of signals: $\{1, 3, 1\} * \{1, 2, 2\}$.
- (c) Multiply the polynomials: $1 + 3z + z^2$ and $1 + 2z + 2z^2$.
- (d) Repeat part (a) for the numbers 1.31 and 12.2.
- (e) Comment on your results.

Question 4

An audio signal $s(t)$ generated by a loudspeaker is reflected at two different walls with reflection coefficients r_1 and r_2 . The signal $x(t)$ recorded by a microphone close to the loudspeaker, after sampling, is

$$x(n) = s(n) + r_1 s(n - k_1) + r_2 s(n - k_2)$$

where k_1 and k_2 are the delays of the two echoes.

- (a) Determine the autocorrelation $r_{xx}(l)$ of the signal $x(n)$.
- (b) Can we obtain r_1 , r_2 , k_1 , and k_2 by observing $r_{xx}(l)$?
- (c) What happens if $r_2 = 0$?

Question 5

Express the output $y(n)$ of a linear time-invariant system with impulse response $h(n)$ in terms of its step response $s(n) = h(n) * u(n)$ and the input $x(n)$.

Question 6

For the sequences

$$x_1(n) = \cos \frac{2\pi}{N}n, \quad x_2(n) = \sin \frac{2\pi}{N}n, \quad 0 \leq n \leq N-1$$

determine the N -point circular convolution $x_1(n) \circledast x_2(n)$

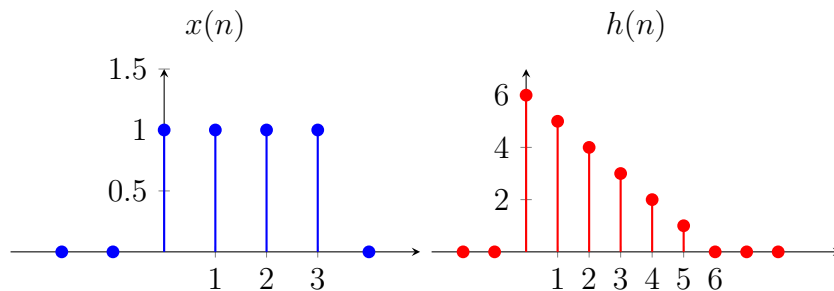
Part II

Coding Questions

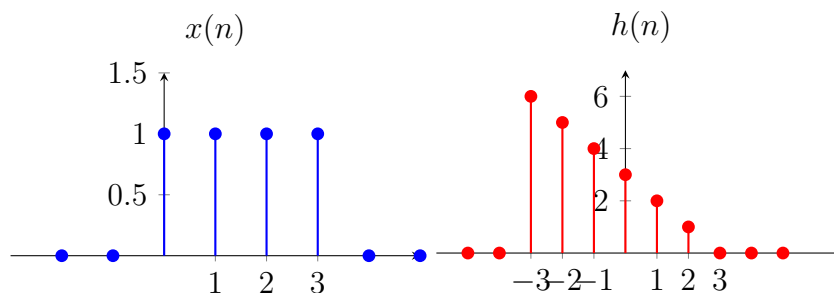
Question 7

Compute and plot the convolutions $x(n) * h(n)$ and $h(n) * x(n)$ for the following pairs of signals.

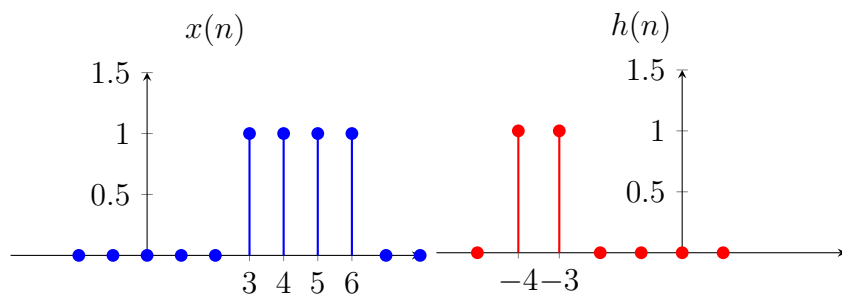
1. Part (a)



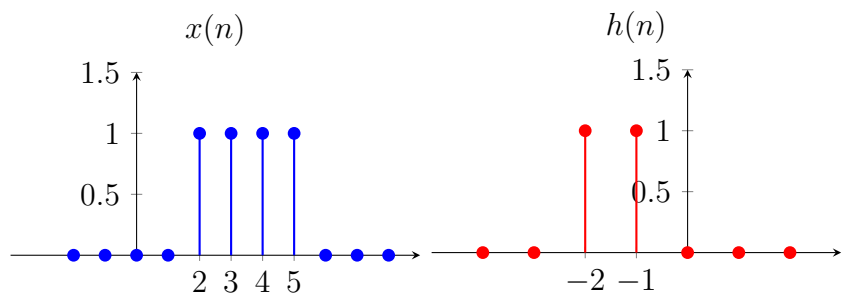
2. Part (b)



3. Part (c)



4. Part (d)



Question 8

Let $x(n]$ be the input signal to a discrete-time filter with impulse response $h_i(n]$ and let $y_i(n]$ be the corresponding output.

$$x(n) = \{1, 4, 2, 3, 5, 3, 3, 4, 5, 7, 6, 9\}$$

$$h_1(n) = \{1, 1\}, \quad h_2(n) = \{1, 2, 1\}, \quad h_3(n) = \left\{\frac{1}{2}, \frac{1}{2}\right\}$$

$$h_4(n) = \left\{\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right\}, \quad h_5(n) = \left\{\frac{1}{4}, -\frac{1}{2}, \frac{1}{4}\right\}$$

- Compute and sketch $x(n]$ and $y_i(n]$ in the following cases, using the same scale in all figures. Sketch $x(n), y_1(n), y_2(n)$ on one graph and $x(n), y_3(n), y_4(n), y_5(n)$ on another graph.
- What is the difference between $y_1(n)$ and $y_2(n)$, and between $y_3(n)$ and $y_4(n)$?
- Comment on the smoothness of $y_2(n)$ and $y_4(n)$. Which factors affect the smoothness?
- Compare $y_4(n)$ with $y_5(n)$. What is the difference? Can you explain it?
- Let $h_6(n) = \left\{\frac{1}{2}, -\frac{1}{2}\right\}$. Compute $y_6(n)$. Sketch $x(n), y_2(n)$, and $y_6(n)$ on the same figure and comment on the results.

Question 9

Use the four-point DFT and IDFT to determine the sequence

$$x_3(n) = x_1(n) \otimes x_2(n)$$

where $x_1(n)$ and $x_2(n)$ are given as:

$$x_1(n) = \{1, 2, 3, 1\}, \quad x_2(n) = \{4, 3, 2, 2\}$$

Question 10

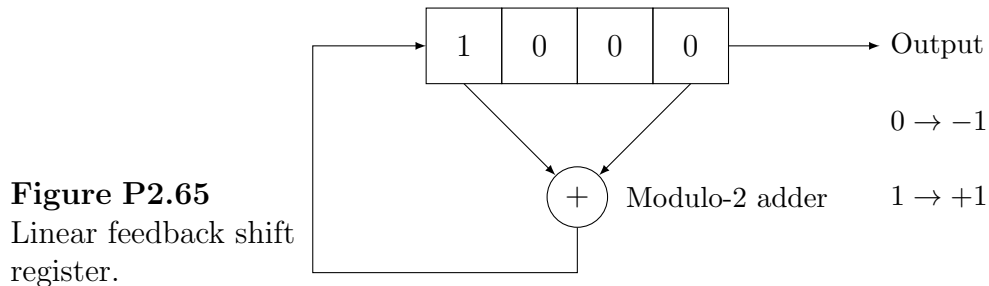
Time-delay estimation in radar

Let $x_a(t)$ be the transmitted signal and $y_a(t)$ be the received signal in a radar system, where

$$y_a(t) = ax_a(t - t_d) + v_a(t)$$

and $v_a(t)$ is additive random noise. The signals $x_a(t)$ and $y_a(t)$ are sampled in the receiver, according to the sampling theorem, and are processed digitally to determine the time delay and hence the distance of the object. The resulting discrete-time signals are

$$\begin{aligned} x(n) &= x_a(nT) \\ y(n) &= y_a(nT) = ax_a(nT - DT) + v_a(nT) \\ &\triangleq ax(n - D) + v(n) \end{aligned}$$



- Explain how we can measure the delay D by computing the crosscorrelation $r_{xv}(l)$.
- Let $x(n)$ be the 13-point *Barker sequence*

$$x(n) = \{+1, +1, +1, +1, +1, -1, -1, +1, +1, -1, +1, -1, +1\}$$

and $v(n)$ be a Gaussian random sequence with zero mean and variance $\sigma^2 = 0.01$. Write a program that generates the sequence $v(n)$, $0 \leq n \leq 199$ for $a = 0.9$ and $D = 20$. Plot the signals $x(n)$, $y(n)$, $0 \leq n \leq 199$.

- Compute and plot the crosscorrelation $r_{xv}(l)$, $0 \leq l \leq 59$. Use the plot to estimate the value of the delay D .

(d) Repeat parts (b) and (c) for $\sigma^2 = 0.1$ and $\sigma^2 = 1$.

(e) Repeat parts (b) and (c) for the signal sequence

$$x(n) = \{ -1, -1, -1, +1, +1, +1, +1, -1, \\ +1, -1, +1, +1, -1, -1, +1 \}$$